GAME MATHEMATICS

TEMPLE DROP

Mathematical model of the published game



96.70%	50,000×	5+	7
RTP in every bet mode	max win per round	cluster pays, board 6 × 6, tumbles	bet modes, 4 purchasable



PREPARED FOR
Limitless Labs

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MODEL VERSION
Math v1 · files published 18 October 2026

DATA BASIS
5,569,743 published rounds and their lookup tables

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Reading conventions

×	Multiple of the base bet (the price of one BASE spin).
Round	One paid spin with all its tumbles and any bonus it triggers, or one purchase. Probabilities are per round.
1 in N	Probability 1/N per round: an average rate, not a schedule.
Hit frequency	Share of rounds that pay more than 0.
Median, Pxx	Half of the rounds pay at most the median; xx% pay at most Pxx (P90, P99…).
Volatility	Standard deviation of the round win per unit staked: the higher, the more the result depends on rare big wins.
RTP points	Percentage of the price returned; the points of a mode add up to 96.70.
Labels	EXACT from the published files MONTE CARLO simulation SPECIFICATION rule or target

CHAPTER 1

Executive summary

7 bet modes, each returning 96.70% of its price, with a 50,000× cap per round.

96.70%
RTP, every mode

50,000×
max win per round

22.0%
hit frequency (BASE), 1 in 4.55

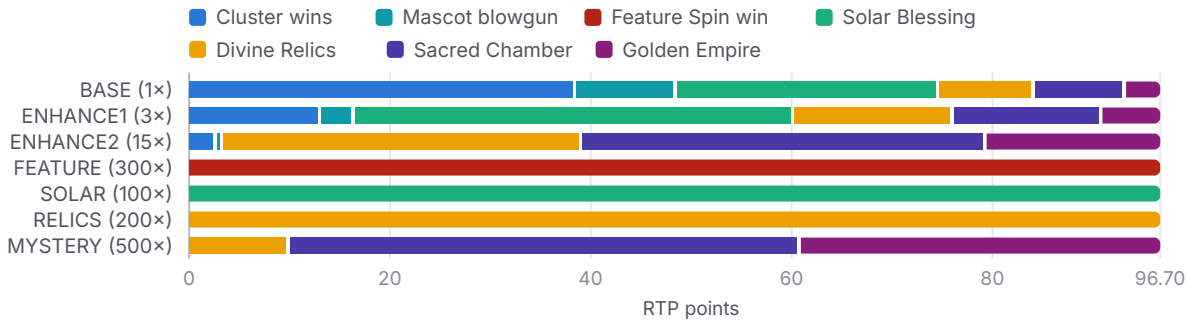
1 in 175
base spins per bonus

Table 1.1 Key indicators by bet mode — EXACT

Mode	Price	RTP	Hit frequency	Win ≥ price	Median win	Volatility (SD / price)	Max win 50,000×
Base game BASE	1×	96.70%	22.0%	1 in 12.5	0×	44.14	1 in 6,587,053
Enhanced Spin 1 ENHANCE1	3×	96.70%	14.0%	1 in 22.0	0×	31.74	1 in 2,014,330
Enhanced Spin 2 ENHANCE2	15×	96.70%	24.8%	1 in 26.3	0×	16.53	1 in 217,958
Feature Spin FEATURE	300×	96.70%	70.0%	1 in 4.89	75.20×	3.34	1 in 30,800
Solar Blessing buy SOLAR	100×	96.70%	100%	1 in 4.58	43.70×	5.65	1 in 40,700
Divine Relics buy RELICS	200×	96.70%	100%	1 in 3.99	80.80×	3.57	1 in 40,700
Mystery Bonus MYSTERY	500×	96.70%	35.0%	1 in 7.29	0×	4.54	1 in 2,159

Per round, exact on the published files. Volatility: standard deviation of the round win per unit staked.

Figure 1.1 Where the return comes from — EXACT



Each bar adds up to 96.70 points. In BASE, the base game brings 50.0% of the return and the bonuses 50.0%.

1.1 Key findings

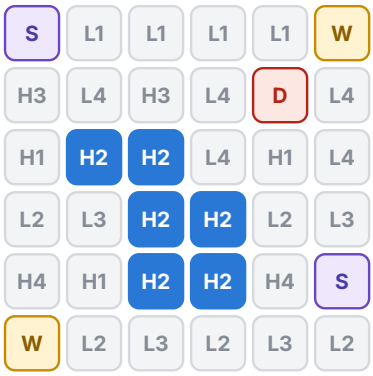
- **Base game** — 22.0% of spins win; a bonus 1 in 175 spins; base game and bonuses share the return 50/50.
- **Mascot** — fires a Dart Blowgun in 1 in 50.0 base spins without a bonus (median 4.80×, mean 5.00×): 10.3% of the BASE return.
- **Golden Empire** — 1 in 250,000 base spins, never below 5,000× (median 7,306×, mean 8,994×).
- **Purchases** — bonus buys: median 40–44% of the price, Divine Relics pays back its price 1 in 3.99; Feature Spin: 30% pay nothing, median 75.20×; Mystery: 65% without a bonus.
- **Verified** — 5,569,743 published rounds re-evaluated: 0 defects; 1 specification item not met: Volatility (brief) (Chapter 14).

CHAPTER 2

Game rules

The rules every published round follows.

2.1 Board, clusters and tumbles



Six reels of six rows. **5 or more** identical symbols connected horizontally or vertically form a **cluster** and pay according to its size; the clusters of a tumble add up.

Golden Shower: after a win, every other cell of a winning symbol becomes a wild. The winning symbols then explode, the symbols above fall and new ones drop in (**tumble**), until no cluster remains.

Example: a 6-cell H2 cluster; the two other H2 turn into wilds (W). D: Dart Blowgun, S: Scatter. Neither pays.

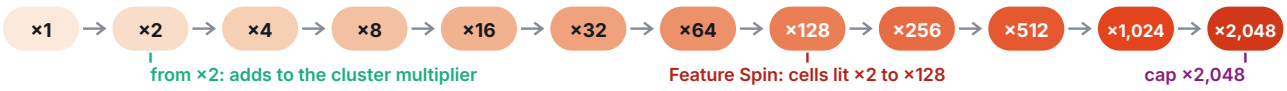
Figure 2.1 Paytable: win of one cluster by size (× base bet) SPECIFICATION

Size →	5	6	7	8	9	10	11	12	13	14	15+
H1 high	0.5×	0.7×	1×	1.5×	2×	3×	4×	5×	7.5×	10×	25×
H2 high	0.4×	0.5×	0.7×	1×	1.5×	2×	3×	4×	5×	7.5×	15×
H3 high	0.3×	0.4×	0.5×	0.7×	1×	1.5×	2×	3×	4×	5×	10×
H4 high	0.2×	0.3×	0.4×	0.5×	0.7×	1×	1.5×	2×	3×	4×	7.5×
L1 low	0.2×	0.2×	0.3×	0.4×	0.5×	0.7×	1×	1.5×	2×	3×	5×
L2 low	0.1×	0.2×	0.2×	0.3×	0.4×	0.5×	0.7×	1×	1.5×	2×	4×
L3 low	0.1×	0.1×	0.2×	0.2×	0.3×	0.4×	0.5×	0.7×	1×	1.5×	3×
L4 low	0.1×	0.1×	0.1×	0.2×	0.2×	0.3×	0.4×	0.5×	0.7×	1×	2×

A cluster win is multiplied by its floor multiplier; a round always ends on a multiple of 0.1× and stops at 50,000×.

2.2 Floor multipliers

Figure 2.2 Value of one floor cell: each explosion on it doubles it SPECIFICATION



Step	Rule
Explosion	the first explosion of a winning symbol on a blank cell marks it ×1; each later explosion on it doubles it, up to ×2,048
Cluster multiplier	sum of the cells at ×2 or more under the cluster (values before the doublings of its own tumble); none = ×1
Temple Totem	doubles every cell of the floor (a blank cell counts ×1 and becomes ×2; cap ×2,048), then becomes a wild
Reset	base game: the floor is cleared at every paid spin; bonus: blank at the start, kept until the end of the bonus

2.3 Special symbols

Symbol	Rule
Wild	substitutes for every paying symbol and counts in every cluster it touches; never on the revealed board: wilds come from the Golden Shower, the darts, and Dart Blowguns and Temple Totems once triggered
Dart Blowgun	the mascot fires 1 to 10 darts at random paying cells, which become wilds; then the blowgun becomes a wild
"?" symbol	lands on the revealed board in a connected group of 5 to 10 cells; all of them turn into the same paying symbol
Scatter	does not pay and never explodes; counted on the final board of the spin, once its tumbles are over
Order	on arrival (revealed board, then new cells of each tumble): "?" first, then Dart Blowguns (reel then row), then Temple Totems

2.4 Free-spin bonuses and bet modes

Tier	Scatters	Free spins	Rule	In BASE	Also
Solar Blessing	3	10	—	1 in 200	ENHANCE1 1/40.0 SOLAR 100×
Divine Relics	4	10	2 Temple Totems on spin 1	1 in 1,500	ENHANCE1 1/300 ENHANCE2 1/33.7 RELICS 200× MYSTERY 23%
Sacred Chamber	5	10	a Dart Blowgun every spin never below 1,000×	1 in 20,000	ENHANCE1 1/4,000 ENHANCE2 1/300 MYSTERY 10%
Golden Empire	6	10	a Dart Blowgun every spin a Temple Totem every spin never below 5,000×	1 in 250,000	ENHANCE1 1/50,000 ENHANCE2 1/3,430 MYSTERY 2%

3 / 4 / 5 / 6 Scatters on a free spin add +10 / +12 / +15 / +20 spins. The floor is kept from one free spin to the next. A bonus ends when its spins run out or at 50,000×.

Mode	Name in the game	Price	What the player gets
`BASE`	Base game	1×	One spin and its tumbles; 3, 4, 5 or 6 Scatters start a bonus.
`ENHANCE1`	Enhanced Spin 1	3×	The same spin; every bonus tier 5.0× as frequent.
`ENHANCE2`	Enhanced Spin 2	15×	The same spin, 4 Scatters or more only: a bonus 1 in 30.0 spins.
`FEATURE`	Feature Spin	300×	One spin on a floor lit at random (×2 to ×128), at least one Dart Blowgun, no Scatter.
`SOLAR`	Solar Blessing buy	100×	Solar Blessing: 10 free spins.
`RELICS`	Divine Relics buy	200×	Divine Relics: 10 free spins, 2 Temple Totems on the first.
`MYSTERY`	Mystery Bonus	500×	65% no bonus · 23% Divine Relics · 10% Sacred Chamber · 2% Golden Empire.

Bonus buys and the Mystery Bonus open on a quiet spin: no win and no special symbol before the bonus.

CHAPTER 3

How the game is published

Pre-computed rounds and their weights: every figure of this document is exact.

Each mode is published as a file of **books** (one complete pre-computed round per line) and a **lookup table** (one integer weight per book). For every round, the game server draws one book with probability weight ÷ total weight. Every per-round figure of this document is therefore a finite weighted sum over the published files: it is computed exactly, not estimated.

Figure 3.1 **Books define what can happen; weights define how often** SPECIFICATION

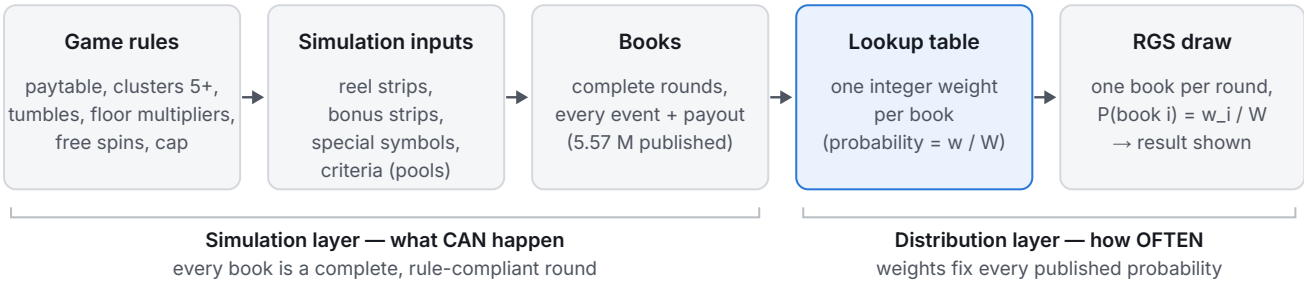


Table 3.1 **Published rounds per mode** EXACT

Mode	Books	Distinct payouts	Max-win books	Most likely single round
BASE	1,275,242	185,722	267,138	1 in 107,406
ENHANCE1	1,275,354	185,902	267,154	1 in 95,705
ENHANCE2	921,689	129,164	193,565	1 in 98,673
FEATURE	499,719	95,886	98,316	1 in 22,375
SOLAR	499,633	135,371	144,964	1 in 45,966
RELICS	498,979	73,211	151,745	1 in 49,203
MYSTERY	599,127	132,608	209,911	1 in 22,709
Total	**5,569,743**			

Rare outcomes are carried by many distinct rounds: no max win is a repeated replay.

Table 3.2 **Nature of the figures**

Label	Meaning
EXACT	Computed on the published lookup tables and books (rounded for display).
MONTE CARLO	Simulated sessions (Sessions chapter only); sample size stated.
SPECIFICATION	Game rules and design targets, compared with the measured values in the last chapter.

CHAPTER 4

The modes compared

Distribution of the round win, relative to the price of the round.

Table 4.1 Percentiles of the round win, in multiples of the price — EXACT

Mode	P25	Median	P75	P90	P99	P99.9	Maximum
BASE	0.00	0.00	0.00	0.30	13.20	97.40	50,000
ENHANCE1	0.00	0.00	0.00	0.03	20.67	115.70	16,667
ENHANCE2	0.00	0.00	0.00	0.07	13.79	130.73	3,333
FEATURE	0.00	0.25	0.72	2.22	13.33	39.44	167
SOLAR	0.19	0.44	0.88	1.85	7.98	31.43	500
RELICS	0.16	0.40	1.00	1.96	7.83	41.26	250
MYSTERY	0.00	0.00	0.30	2.35	17.63	71.81	100

Read: in BASE, 90% of spins pay at most the P90 value (× the price). 0 = at least that share of rounds pays nothing.

Table 4.2 Chance per round of winning at least k times the price — EXACT

Mode	≥ 0.5×	≥ 1×	≥ 2×	≥ 5×	≥ 10×	≥ 20×	≥ 50×	≥ 100×
BASE	8.97% 1 in 11.1	8.02% 1 in 12.5	3.42% 1 in 29.2	2.21% 1 in 45.2	1.15% 1 in 86.9	0.715% 1 in 140	0.315% 1 in 317	0.0957% 1 in 1,045
ENHANCE1	5.37% 1 in 18.6	4.54% 1 in 22.0	3.35% 1 in 29.8	2.35% 1 in 42.5	1.89% 1 in 53.0	1.06% 1 in 94.7	0.248% 1 in 403	0.118% 1 in 849
ENHANCE2	4.18% 1 in 23.9	3.80% 1 in 26.3	3.52% 1 in 28.4	2.74% 1 in 36.4	1.63% 1 in 61.2	0.696% 1 in 144	0.411% 1 in 243	0.184% 1 in 544
FEATURE	32.4% 1 in 3.08	20.5% 1 in 4.89	11.3% 1 in 8.82	3.30% 1 in 30.3	1.35% 1 in 74.0	0.514% 1 in 194	0.0531% 1 in 1,882	0.0136% 1 in 7,355
SOLAR	44.0% 1 in 2.27	21.9% 1 in 4.58	8.72% 1 in 11.5	2.31% 1 in 43.3	0.623% 1 in 161	0.179% 1 in 559	0.0621% 1 in 1,611	0.0313% 1 in 3,194
RELICS	41.8% 1 in 2.39	25.0% 1 in 3.99	9.68% 1 in 10.3	1.97% 1 in 50.7	0.708% 1 in 141	0.263% 1 in 380	0.0650% 1 in 1,538	0.0209% 1 in 4,788
MYSTERY	17.1% 1 in 5.86	13.7% 1 in 7.29	12.1% 1 in 8.24	3.57% 1 in 28.0	2.55% 1 in 39.2	0.797% 1 in 125	0.172% 1 in 581	0.0463% 1 in 2,159

"—": above the 50,000× cap for this mode.

Figure 4.1 How often one round pays at least... (× the base bet) EXACT

	100×	1,000×	10,000×	50,000× (max win)
BASE 1×	1 in 1,045	1 in 12,891	1 in 531,534	1 in 6.6 million
ENHANCE1 3×	1 in 221	1 in 3,020	1 in 106,202	1 in 2.0 million
ENHANCE2 15×	1 in 40	1 in 259	1 in 11,299	1 in 217,958
FEATURE 300×	1 in 2.3	1 in 19	1 in 681	1 in 30,800
SOLAR 100×	1 in 4.6	1 in 161	1 in 3,194	1 in 40,700
RELICS 200×	1 in 2.4	1 in 51	1 in 1,538	1 in 40,700
MYSTERY 500×	1 in 3.5	1 in 8.2	1 in 125	1 in 2,159

Per round of the mode; darker = more frequent.

Table 4.3 Volatility against the brief SPECIFICATION

Mode	Price	Brief rating	Target SD / price	Published SD / price	Status
ENHANCE1	3×	4/5	30.90 – 34.63	31.74	met
FEATURE	300×	4/5	3.60 – 4.64	3.34	not met
SOLAR	100×	5/5	≥ 5.51	5.65	met
RELICS	200×	4/5	3.71 – 5.23	3.57	not met
MYSTERY	500×	5/5	≥ 4.31	4.54	met

The brief rates the volatility of each mode out of 5. Each rating is a design target on the SD / price (standard deviation of the round win divided by the price of the mode), set for the price of the mode.

CHAPTER 5

Base game (BASE)

1× per round — One spin and its tumbles; 3, 4, 5 or 6 Scatters start a bonus.

Return to player	Hit frequency	Win ≥ price	Median win
96.70%	**22.0%**	**8.02%**	**0×**
mean win 0.97×	1 in 4.55	1 in 12.5	0% of the price

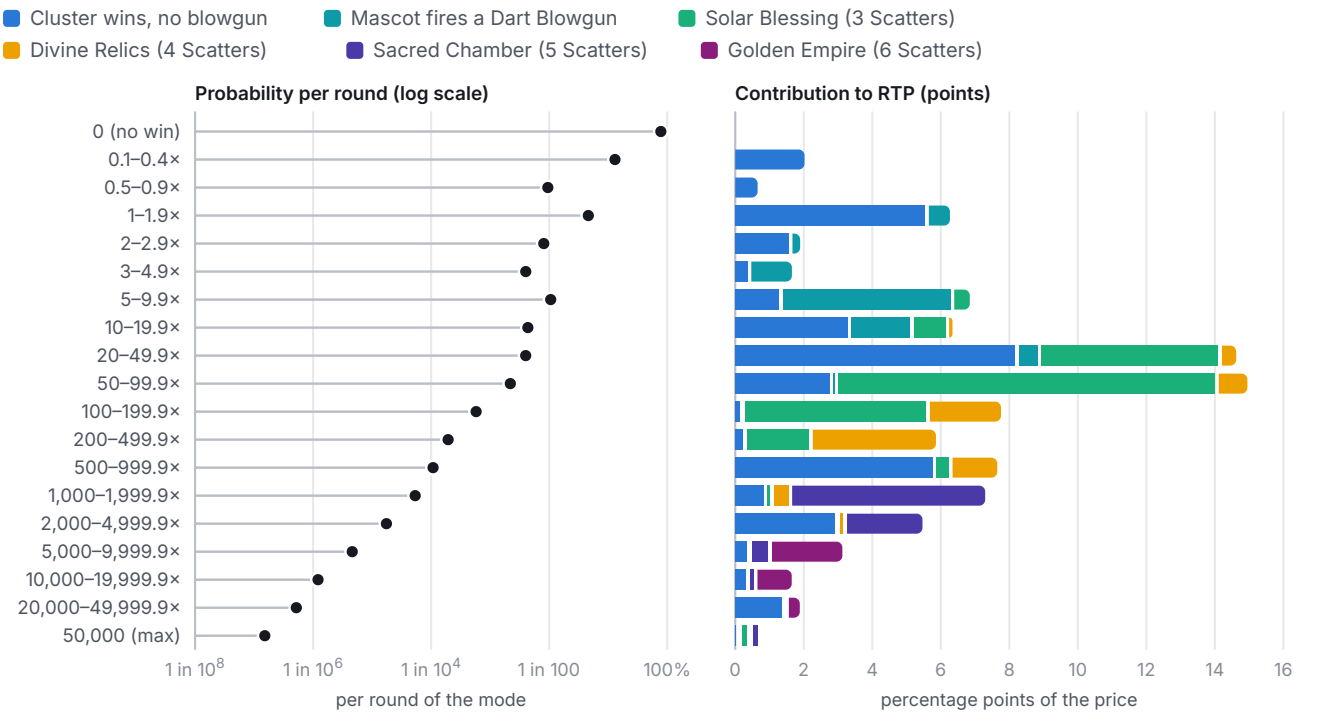
Volatility	P90 / P99	Max win 50,000×	Published rounds
44.14	**0.30× / 13.20×**	**1 in 6,587,053**	**1,275,242**
SD per unit staked	90% / 99% of rounds pay at most	1.52×10^{-5}%	185,722 distinct payouts

Table 5.1 Outcome classes — EXACT

Class	Probability	Frequency	Share of RTP	Mean win	Median	P10 – P90
No win	77.9%	1 in 1.28	0.0%	—	—	—
Cluster wins, no blowgun	19.6%	1 in 5.11	39.7%	1.96×	0.20×	0.10× – 1.70×
Mascot fires a Dart Blowgun	2.00%	1 in 50.0	10.3%	5.00×	4.80×	1.00× – 9.30×
Solar Blessing (3 Scatters)	0.500%	1 in 200	27.1%	52.36×	43.10×	8.10× – 99.00×
Divine Relics (4 Scatters)	0.0667%	1 in 1,500	9.8%	142.7×	82.20×	15.20× – 333.2×
Sacred Chamber (5 Scatters)	0.00500%	1 in 20,000	9.4%	1,811.0×	1,460.4×	1,089.5× – 2,609.9×
Golden Empire (6 Scatters)	0.000400%	1 in 250,000	3.7%	8,993.6×	7,306.4×	5,464.4× – 14,032.8×

Every round belongs to one class; mean, median and P10–P90 are per round of the class.

Figure 5.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 6

Enhanced Spin 1 (ENHANCE1)

3× per round — The same spin; every bonus tier 5.0× as frequent.

Return to player	Hit frequency	Win ≥ price	Median win
96.70%	**14.0%**	**4.54%**	**0×**
mean win 2.90×	1 in 7.14	1 in 22.0	0% of the price

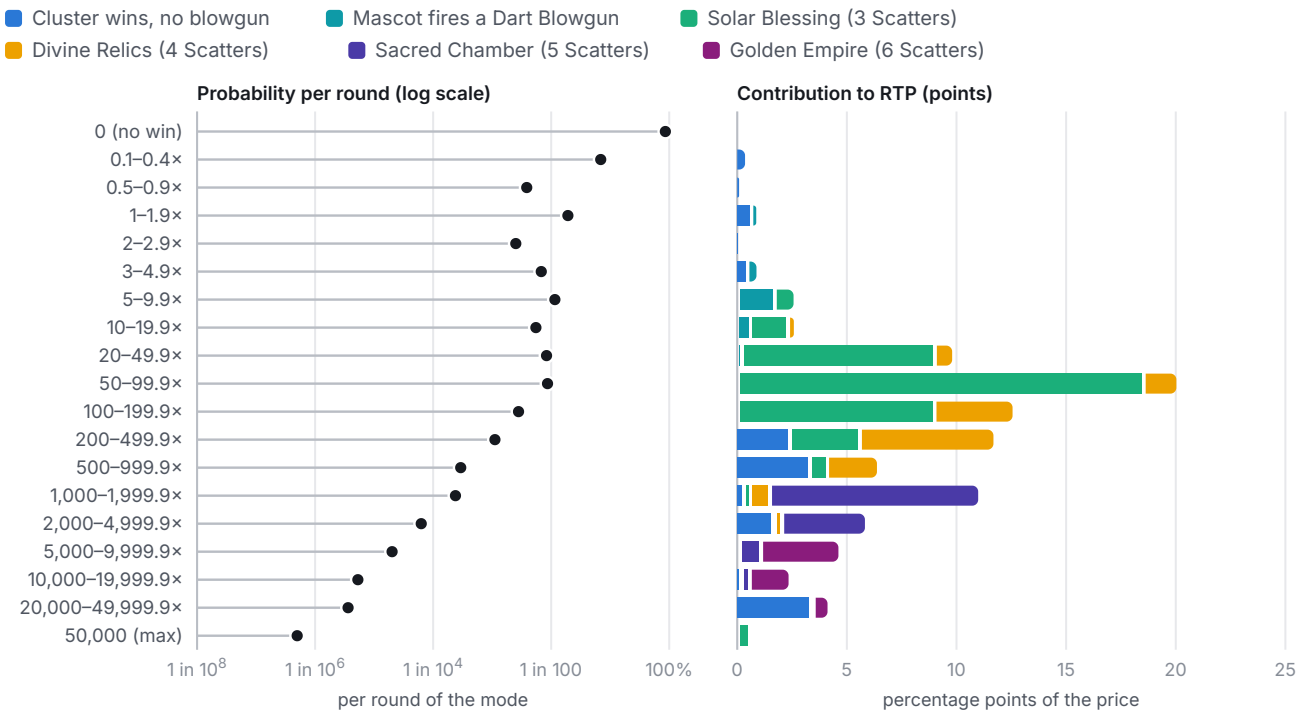
Volatility	P90 / P99	Max win 50,000×	Published rounds
31.74	**0.10× / 62.00×**	**1 in 2,014,330**	**1,275,354**
SD per unit staked	90% / 99% of rounds pay at most	4.96×10^{-5}%	185,902 distinct payouts

Table 6.1 Outcome classes — EXACT

Class	Probability	Frequency	Share of RTP	Mean win	Median	P10 – P90
No win	85.9%	1 in 1.16	0.0%	—	—	—
Cluster wins, no blowgun	9.29%	1 in 10.8	13.4%	4.19×	0.20×	0.10× – 1.50×
Mascot fires a Dart Blowgun	2.00%	1 in 50.0	3.4%	5.00×	4.70×	1.00× – 9.30×
Solar Blessing (3 Scatters)	2.50%	1 in 40.0	45.2%	52.49×	43.10×	8.10× – 99.10×
Divine Relics (4 Scatters)	0.333%	1 in 300	16.4%	142.8×	82.10×	15.20× – 331.3×
Sacred Chamber (5 Scatters)	0.0250%	1 in 4,000	15.3%	1,778.6×	1,460×	1,089× – 2,602.2×
Golden Empire (6 Scatters)	0.00200%	1 in 50,000	6.2%	8,932.0×	7,302.6×	5,464× – 13,990.6×

Every round belongs to one class; mean, median and P10–P90 are per round of the class.

Figure 6.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 7

Enhanced Spin 2 (ENHANCE2)

15× per round — The same spin, 4 Scatters or more only: a bonus 1 in 30.0 spins.

Return to player	Hit frequency	Win ≥ price	Median win
96.70%	**24.8%**	**3.80%**	**0×**
mean win 14.50×	1 in 4.04	1 in 26.3	0% of the price

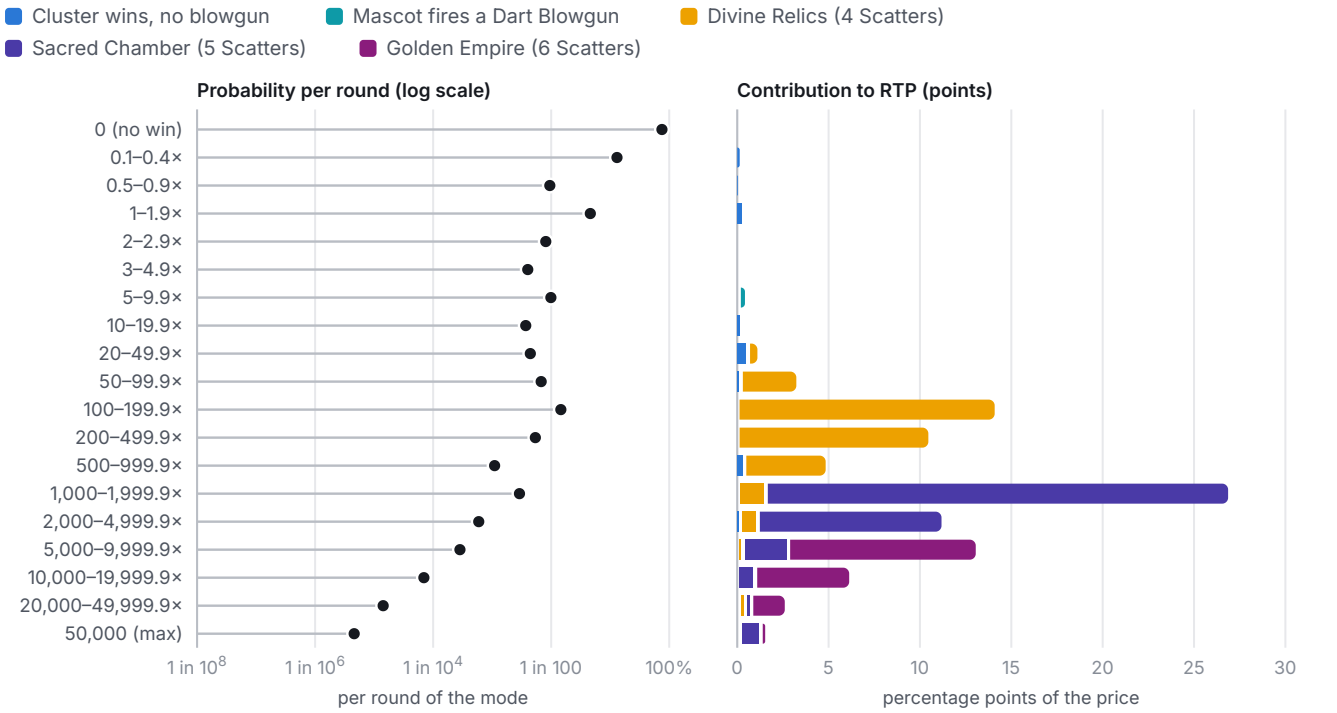
Volatility	P90 / P99	Max win 50,000×	Published rounds
16.53	**1.10× / 206.8×**	**1 in 217,958**	**921,689**
SD per unit staked	90% / 99% of rounds pay at most	0.000459%	129,164 distinct payouts

Table 7.1 Outcome classes — EXACT

Class	Probability	Frequency	Share of RTP	Mean win	Median	P10 – P90
No win	75.1%	1 in 1.33	0.0%	—	—	—
Cluster wins, no blowgun	19.6%	1 in 5.11	2.6%	1.96×	0.20×	0.10× – 1.70×
Mascot fires a Dart Blowgun	2.00%	1 in 50.0	0.7%	5.00×	4.70×	1.00× – 9.30×
Divine Relics (4 Scatters)	2.97%	1 in 33.7	37.0%	180.5×	138.4×	55.20× – 315.6×
Sacred Chamber (5 Scatters)	0.333%	1 in 300	41.6%	1,810.9×	1,460.2×	1,088.1× – 2,610.9×
Golden Empire (6 Scatters)	0.0292%	1 in 3,430	18.1%	8,995.7×	7,308.8×	5,469.6× – 14,012.8×

Every round belongs to one class; mean, median and P10–P90 are per round of the class.

Figure 7.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 8

Feature Spin (FEATURE)

300× per round — One spin on a floor lit at random (×2 to ×128), at least one Dart Blowgun, no Scatter.

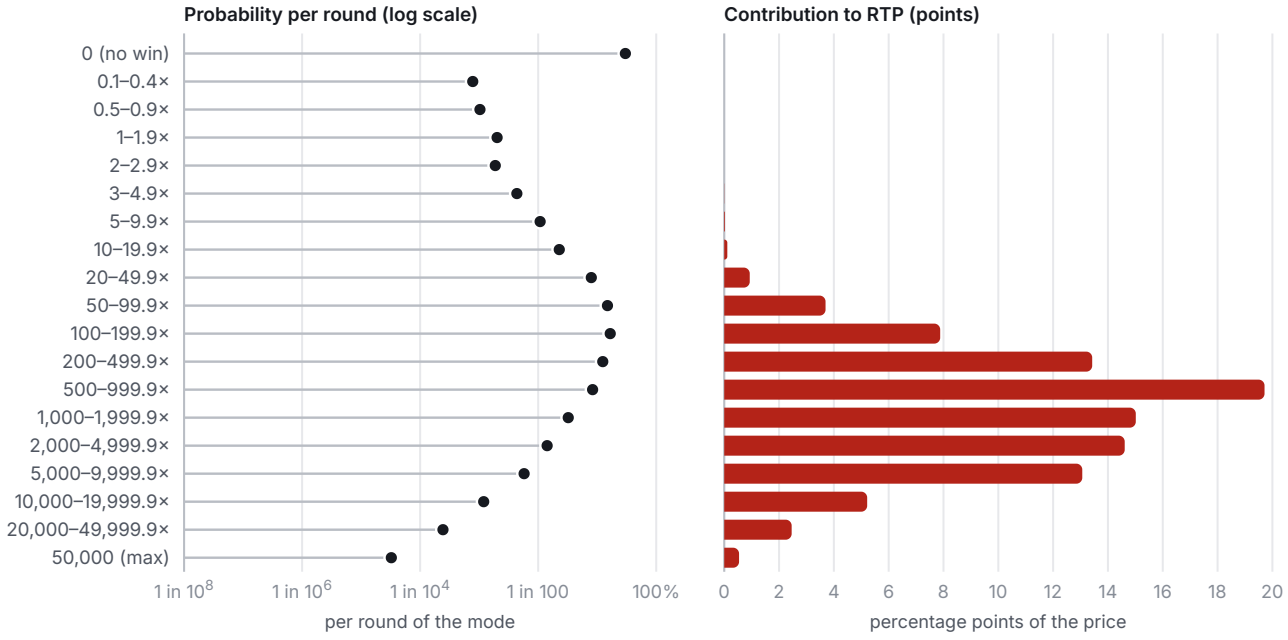
Return to player	Hit frequency	Win ≥ price	Median win
96.70%	**70.0%**	**20.5%**	**75.20×**
mean win 290.1×	1 in 1.43	1 in 4.89	25% of the price

Volatility	P90 / P99	Max win 50,000×	Published rounds
3.34	**664.8× / 4,000.2×**	**1 in 30,800**	**499,719**
SD per unit staked	90% / 99% of rounds pay at most	0.00325%	95,886 distinct payouts

Table 8.1 Result relative to the price — EXACT

Round win	Probability	Frequency
Nothing	30.0%	1 in 3.33
Below ½ the price	37.6%	1 in 2.66
0.5× to 1× the price	12.0%	1 in 8.34
1× to 2× the price	9.12%	1 in 11.0
2× to 5× the price	8.04%	1 in 12.4
5× to 10× the price	1.95%	1 in 51.4
10× to 20× the price	0.838%	1 in 119
20× the price or more	0.514%	1 in 194

Figure 8.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 9

Solar Blessing buy (SOLAR)

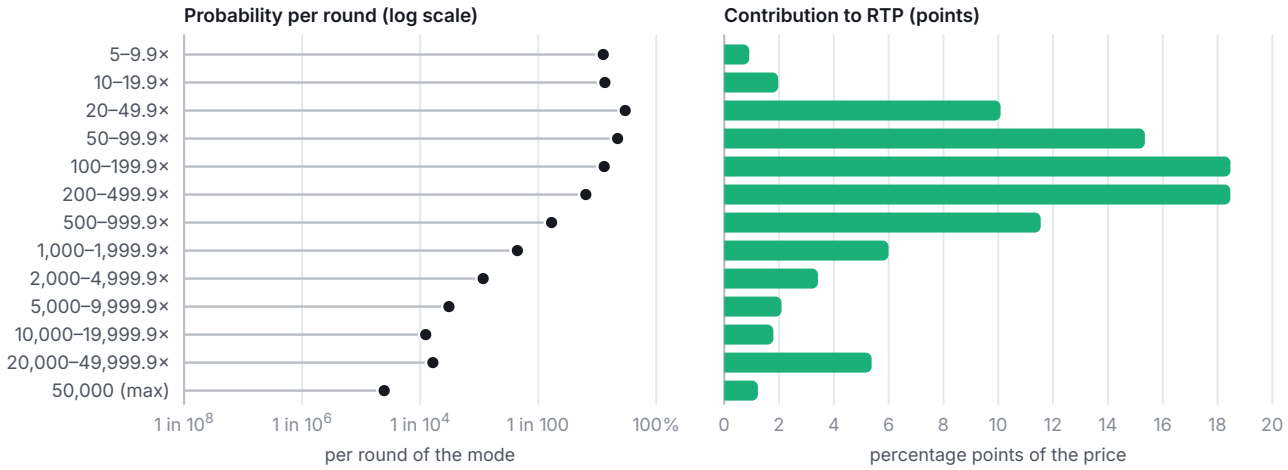
100× per round — Solar Blessing: 10 free spins.

Return to player **96.70%** mean win 96.70×	Hit frequency **100%** every round pays	Win ≥ price **21.9%** 1 in 4.58	Median win **43.70×** 44% of the price
Volatility **5.65** SD per unit staked	P90 / P99 **185.1× / 797.9×** 90% / 99% of rounds pay at most	Max win 50,000× **1 in 40,700** 0.00246%	Published rounds **499,633** 135,371 distinct payouts

Table 9.1 Result relative to the price — EXACT

Round win	Probability	Frequency
Nothing	0%	—
Below ½ the price	56.0%	1 in 1.79
0.5× to 1× the price	22.2%	1 in 4.51
1× to 2× the price	13.1%	1 in 7.62
2× to 5× the price	6.41%	1 in 15.6
5× to 10× the price	1.69%	1 in 59.3
10× to 20× the price	0.444%	1 in 225
20× the price or more	0.179%	1 in 559

Figure 9.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 10

Divine Relics buy (RELICS)

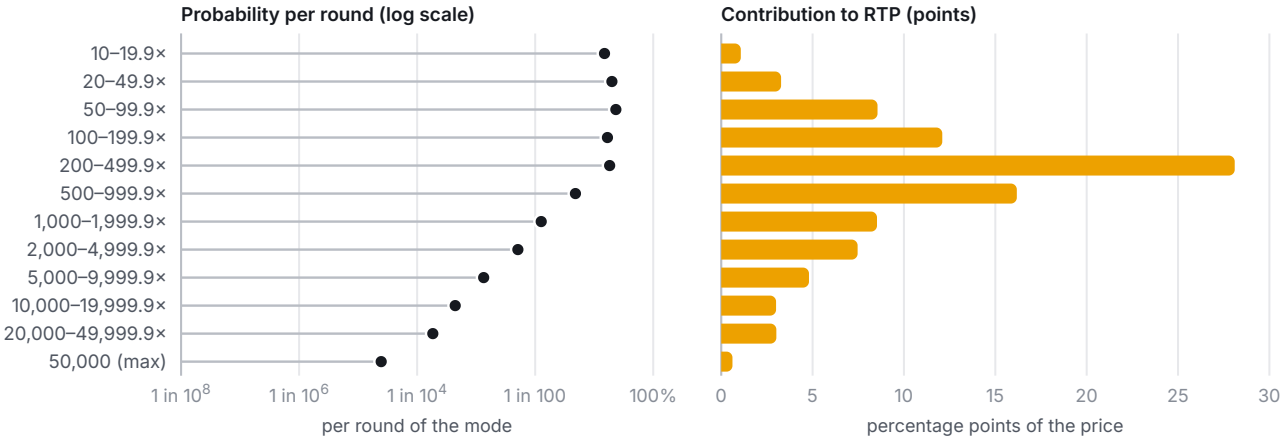
200× per round — Divine Relics: 10 free spins, 2 Temple Totems on the first.

Return to player **96.70%** mean win 193.4×	Hit frequency **100%** every round pays	Win ≥ price **25.0%** 1 in 3.99	Median win **80.80×** 40% of the price
Volatility **3.57** SD per unit staked	P90 / P99 **392.4× / 1,565.6×** 90% / 99% of rounds pay at most	Max win 50,000× **1 in 40,700** 0.00246%	Published rounds **498,979** 73,211 distinct payouts

Table 10.1 Result relative to the price — EXACT

Round win	Probability	Frequency
Nothing	0%	—
Below ½ the price	58.2%	1 in 1.72
0.5× to 1× the price	16.8%	1 in 5.96
1× to 2× the price	15.4%	1 in 6.50
2× to 5× the price	7.70%	1 in 13.0
5× to 10× the price	1.27%	1 in 79.0
10× to 20× the price	0.445%	1 in 225
20× the price or more	0.263%	1 in 380

Figure 10.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 11

Mystery Bonus (MYSTERY)

500× per round — 65% no bonus · 23% Divine Relics · 10% Sacred Chamber · 2% Golden Empire.

Return to player	Hit frequency	Win ≥ price	Median win
96.70%	**35.0%**	**13.7%**	**0×**
mean win 483.5×	1 in 2.86	1 in 7.29	0% of the price

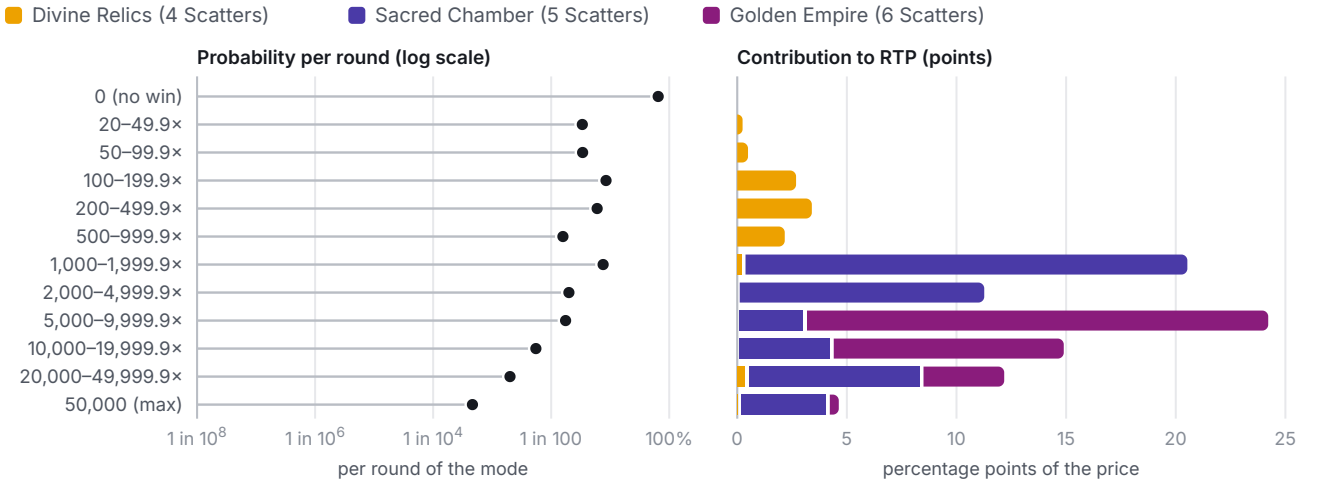
Volatility	P90 / P99	Max win 50,000×	Published rounds
4.54	**1,177.4× / 8,812.8×**	**1 in 2,159**	**599,127**
SD per unit staked	90% / 99% of rounds pay at most	0.0463%	132,608 distinct payouts

Table 11.1 Outcome classes — EXACT

Class	Probability	Frequency	Share of RTP	Mean win	Median	P10 – P90
No bonus (Mystery)	65.0%	1 in 1.54	0.0%	—	—	—
Divine Relics (4 Scatters)	23.0%	1 in 4.35	10.2%	214.1×	166.8×	40.00× – 409.6×
Sacred Chamber (5 Scatters)	10.0%	1 in 10.00	52.6%	2,543.5×	1,433.8×	1,097.1× – 3,339.9×
Golden Empire (6 Scatters)	2.00%	1 in 50.0	37.2%	8,995.1×	7,304.8×	5,468× – 14,052×

Every round belongs to one class; mean, median and P10–P90 are per round of the class.

Figure 11.1 Payout distribution: probability and RTP contribution of each band (× base bet) — EXACT



Left: chance that a round ends in the band (log scale). Right: the band's share of the RTP, by outcome class.

CHAPTER 12

The free-spin bonuses

Each tier as triggered in the base game: frequency, length, mechanics and win.

Table 12.1 The four tiers, as triggered in the base game — EXACT

Tier	In BASE	In ENHANCE1	Spins	Retriggered	Mean win	Median	P10 – P90	Max win
Solar Blessing	1 in 200	1 in 40.0	10 → 10.7	6.7%	52.36×	43.10×	8.10× – 99.00×	1 in 100,000
Divine Relics	1 in 1,500	1 in 300	10 → 10.5	4.4%	142.7×	82.20×	15.20× – 333.2×	1 in 59,999
Sacred Chamber	1 in 20,000	1 in 4,000	10 → 10.9	8.9%	1,811.0×	1,460.4×	1,089.5× – 2,609.9×	1 in 1,000
Golden Empire	1 in 250,000	1 in 50,000	10 → 10.0	0.1%	8,993.6×	7,306.4×	5,464.4× – 14,032.8×	1 in 400

Spins: awarded → played on average, retriggers included. Mean, median, P10–P90 and max win are per bonus (× base bet), the trigger spin included.

Table 12.2 What happens during a bonus (average per bonus, as triggered in BASE) — EXACT

Tier	Dart Blowguns	Darts → wilds	Temple Totems	Golden Shower wilds	Highest floor cell (median)	Floor reaches ×2,048
Solar Blessing	1.2	4.3	0.5	21.4	×8	0.0%
Divine Relics	1.0	3.3	2.3	10.5	×16	0.0%
Sacred Chamber	10.9	17.9	0.3	61.2	×128	1.6%
Golden Empire	10.0	15.6	10.0	28.8	×2,048	99.9%

Counted in the published books of each bonus (all its free spins). Highest floor cell: the largest floor value reached during the bonus.

Figure 12.1 Win of a bonus: P10 – P90 range, median (dot) and mean (line), log scale — EXACT

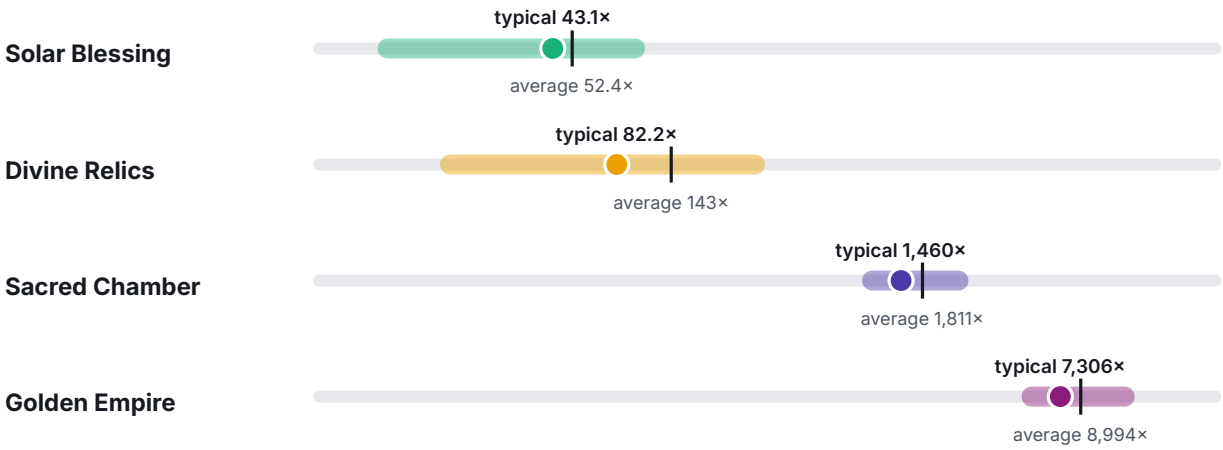


Table 12.3 Bought (or Mystery) bonuses against the same tier triggered in BASE — EXACT

Mode	Tier	Mean	Mean in BASE	Median	Median in BASE	Lowest win
SOLAR	Solar Blessing	96.70×	52.36×	43.70×	43.10×	5.0×
RELICS	Divine Relics	193.4×	142.7×	80.80×	82.20×	10.0×
MYSTERY	Divine Relics	214.1×	142.7×	166.8×	82.20×	25.2×
MYSTERY	Sacred Chamber	2,543.5×	1,811.0×	1,433.8×	1,460.4×	1,000.1×
MYSTERY	Golden Empire	8,995.1×	8,993.6×	7,304.8×	7,306.4×	5,000.4×

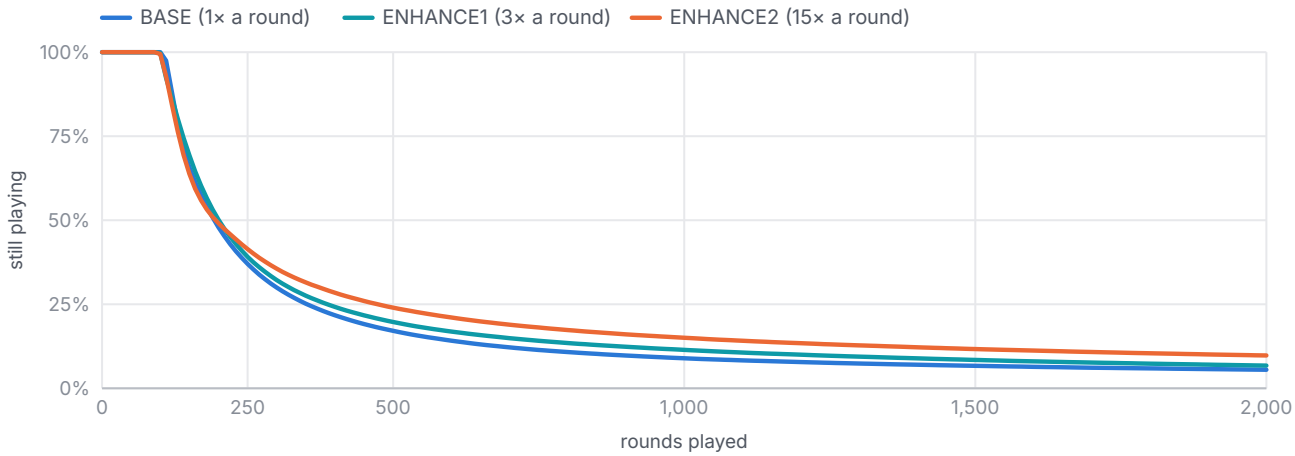
CHAPTER 13

Playing sessions

What many rounds in a row look like: credit life, results, big wins.

Figure 13.1 Players still playing, starting with a credit of 100 rounds

MONTE CARLO



200,000 simulated players per curve; a player stops when the credit no longer covers a round.

Table 13.1 Results of a session

MONTE CARLO

Mode	Rounds	Ahead or even	Median return	Hit a big win	Big win =
BASE	100	22.3%	49%	9.1%	100×
BASE	1,000	23.9%	71%	61.6%	100×
BASE	5,000	28.8%	84%	99.2%	100×
ENHANCE1	100	22.6%	53%	11.1%	300×
ENHANCE1	1,000	28.4%	76%	69.2%	300×
ENHANCE1	5,000	28.8%	86%	99.7%	300×
ENHANCE2	100	30.0%	47%	16.9%	1,500×
ENHANCE2	1,000	35.6%	84%	84.1%	1,500×
ENHANCE2	5,000	37.3%	93%	100.0%	1,500×
FEATURE	10	29.2%	68%	12.7%	3,000×
FEATURE	100	37.1%	90%	74.3%	3,000×
SOLAR	10	26.5%	72%	6.0%	1,000×
SOLAR	100	23.6%	85%	46.4%	1,000×
RELICS	10	27.7%	75%	6.8%	2,000×
RELICS	100	30.0%	88%	50.9%	2,000×
MYSTERY	10	28.0%	48%	22.7%	5,000×
MYSTERY	100	38.0%	87%	92.4%	5,000×

Return: total won ÷ total staked over the session. Big win: at least one round paying 100× the price of a spin, or 10× the price of a purchase. 1,000,000 simulated sessions per line.

CHAPTER 14

Verification

Every published round re-evaluated; every design target compared with the published game.

5,569,743
rounds re-evaluated by a second program, separate from the game engine

0
defects found

0
books inconsistent with their events or their lookup-table row

15 / 16
specification items met

The re-evaluation replays every round from its board and events with the game rules — clusters, Golden Shower, floor multipliers, Dart Blowguns, Temple Totems, "?" groups, tumbles drawn from the reel strips, bonuses and their guarantees, retriggers, cap — and checks its payout against the lookup table. The RTP of each mode is the exact weighted sum over its lookup table.

Table 14.1 Design targets against the published game — SPECIFICATION

Item	Target	Published game	Status
RTP of every mode	96.70%	96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000%	met
Maximum win	50,000× in every mode	50,000× / 50,000× / 50,000× / 50,000× / 50,000× / 50,000× / 50,000×	met
BASE bonus frequencies	3S 1/200 · 4S 1/1,500 · 5S 1/20,000 · 6S 1/250,000	1 in 200 · 1 in 1,500 · 1 in 20,000 · 1 in 250,000	met
Enhanced Spin 1	every bonus tier 5× as frequent as in BASE	×5.00 · ×5.00 · ×5.00 · ×5.00	met
Enhanced Spin 2	4 Scatters or more only; a bonus 1 in 30; 5S 1/300, 6S 1/3,430	no 3-Scatter bonus; a bonus 1 in 30.0; 5S 1 in 300, 6S 1 in 3,430	met
Tiers ordered	more Scatters: rarer and richer on average	52× < 143× < 1,811× < 8,994×	met
Bonus floors	Solar Blessing ≥ 5× · Divine Relics ≥ 10× · Sacred Chamber ≥ 1,000× · Golden Empire ≥ 5,000×	5.0× · 10.0× · 1,000.1× · 5,000.1×	met
Mascot	fires about 1 in 50 BASE spins; when it pays, at least the bet	1 in 50.0; lowest paying 1.0×	met
Feature Spin	30% of spins pay nothing; at least one Dart Blowgun; no Scatter	30.00% pay nothing; spins without a Dart Blowgun 0%; no bonus	met
Bonus buys	median ≈ 45% of the price; Divine Relics pays back its price 1 time in 4	medians 44% / 40%; Divine Relics 25.0% (1 in 3.99)	met
Mystery mix	65% / 23% / 10% / 2% (no bonus / Divine Relics / Sacred Chamber / Golden Empire)	65.00% / 23.00% / 10.00% / 2.000%	met
Volatility (brief)	ENHANCE1 4/5 · FEATURE 4/5 · SOLAR 5/5 · RELICS 4/5 · MYSTERY 5/5	ENHANCE1 31.74 · FEATURE 3.34 · SOLAR 5.65 · RELICS 3.57 · MYSTERY 4.54	not met
Stake 3-star tail limits (every mode)	P(≥ 5,000×) ≤ 5%, P(≥ 10,000×) ≤ 1%, P(≥ 25,000×) ≤ 0.5%; wins above 10,000× carry ≤ 80% of the RTP; highest price × 1,000 ≤ 500,000	largest: 2.55% / 0.797% / 0.172%; 32.8%; 500,000	met
Book files	each book file under 4.2 GB	largest 3.06 GB	met
Pools	≥ 100,000 books per bonus tier and per feature; ≥ 10,000 max-win books per mode	smallest pool 126,841 books; max-win books 267,138 / 267,154 / 193,565 / 98,316 / 144,964 / 151,745 / 209,911	met
Rule compliance	every book follows the rules of the game	0 defects (re-evaluation of every round); 0 event inconsistencies	met